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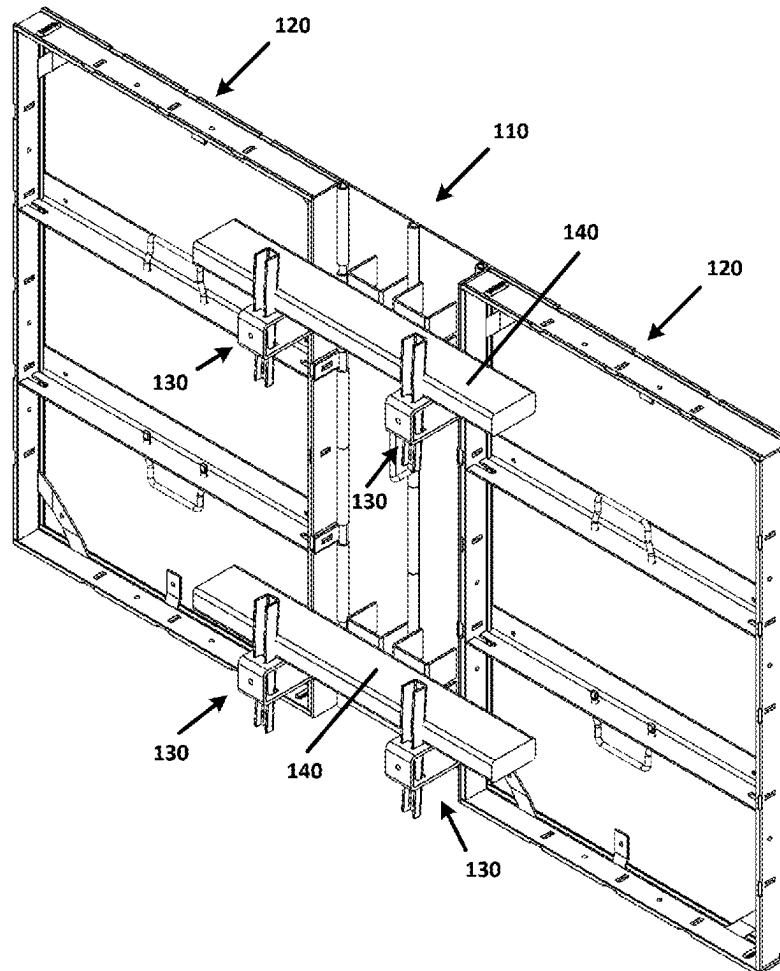
(19) **United States**(12) **Patent Application Publication**
Weaver et al.(10) **Pub. No.: US 2025/0277380 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **COLLAPSIBLE FORM**(52) **U.S. Cl.**CPC *E04G 17/04* (2013.01); *E04G 17/14*
(2013.01)(71) Applicant: **Edmar Manufacturing, Inc.**, Holland,
MI (US)(72) Inventors: **Richard Weaver**, Allegan, MI (US);
Derrick Richardson, Fennville, MI
(US)

(57)

ABSTRACT(21) Appl. No.: **19/065,390**(22) Filed: **Feb. 27, 2025****Related U.S. Application Data**(60) Provisional application No. 63/560,606, filed on Mar.
1, 2024.**Publication Classification**(51) **Int. Cl.***E04G 17/04* (2006.01)*E04G 17/14* (2006.01)

A collapsible concrete form includes a pair of inner panels hingedly coupled to one another and a pair of outer flanges configured to abut a pair of adjacent form sections. A first one of the pair of outer flanges is hingedly coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges is hingedly coupled to a second one of the pair of inner panels. The collapsible concrete form further includes a handle protruding from one of the pair of inner panels or a hinge coupling the pair of inner panels. The handle is operable to move the pair of inner panels and the pair of outer flanges between a deployed arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner panels are askew.

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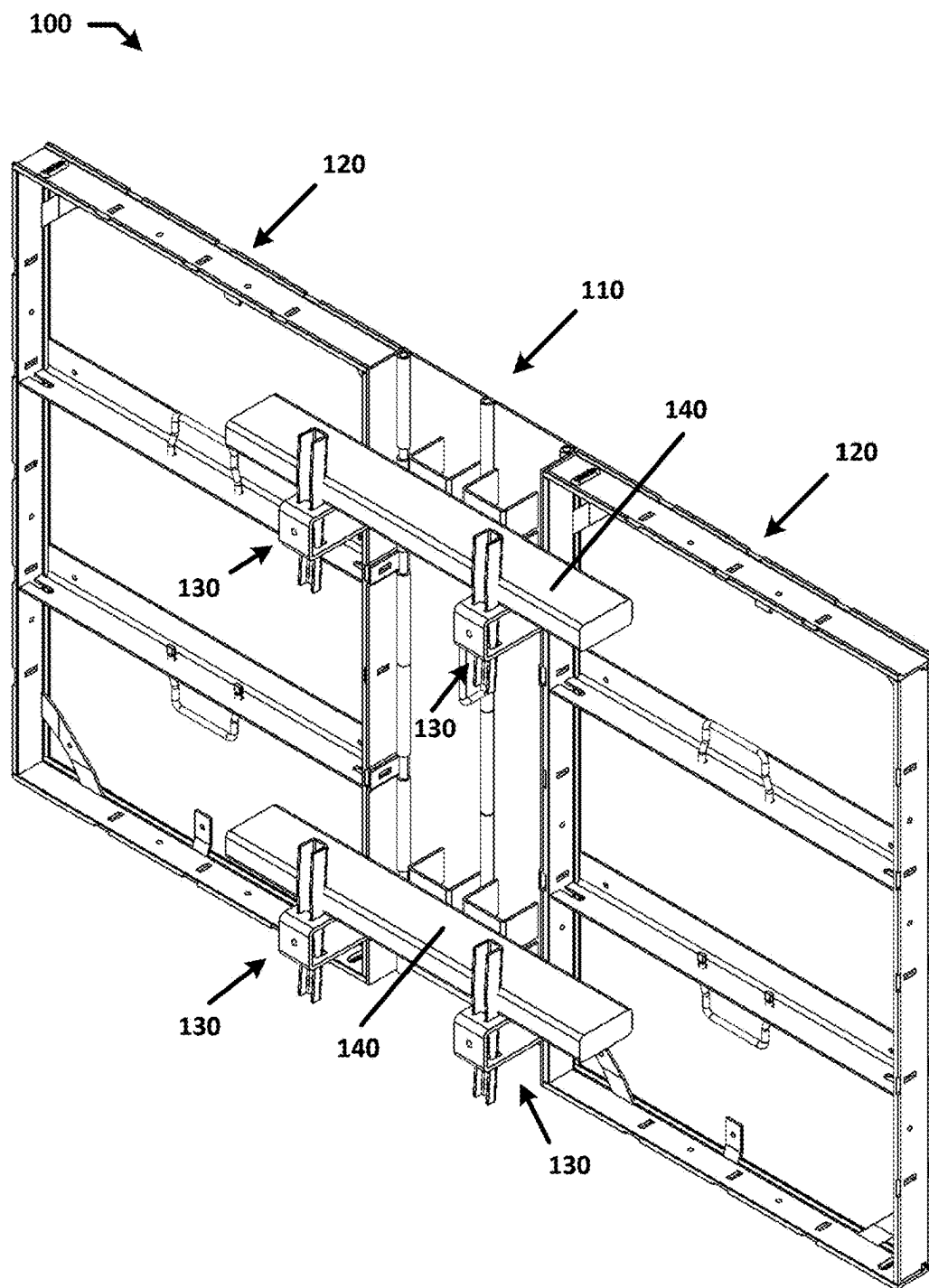


FIG. 1

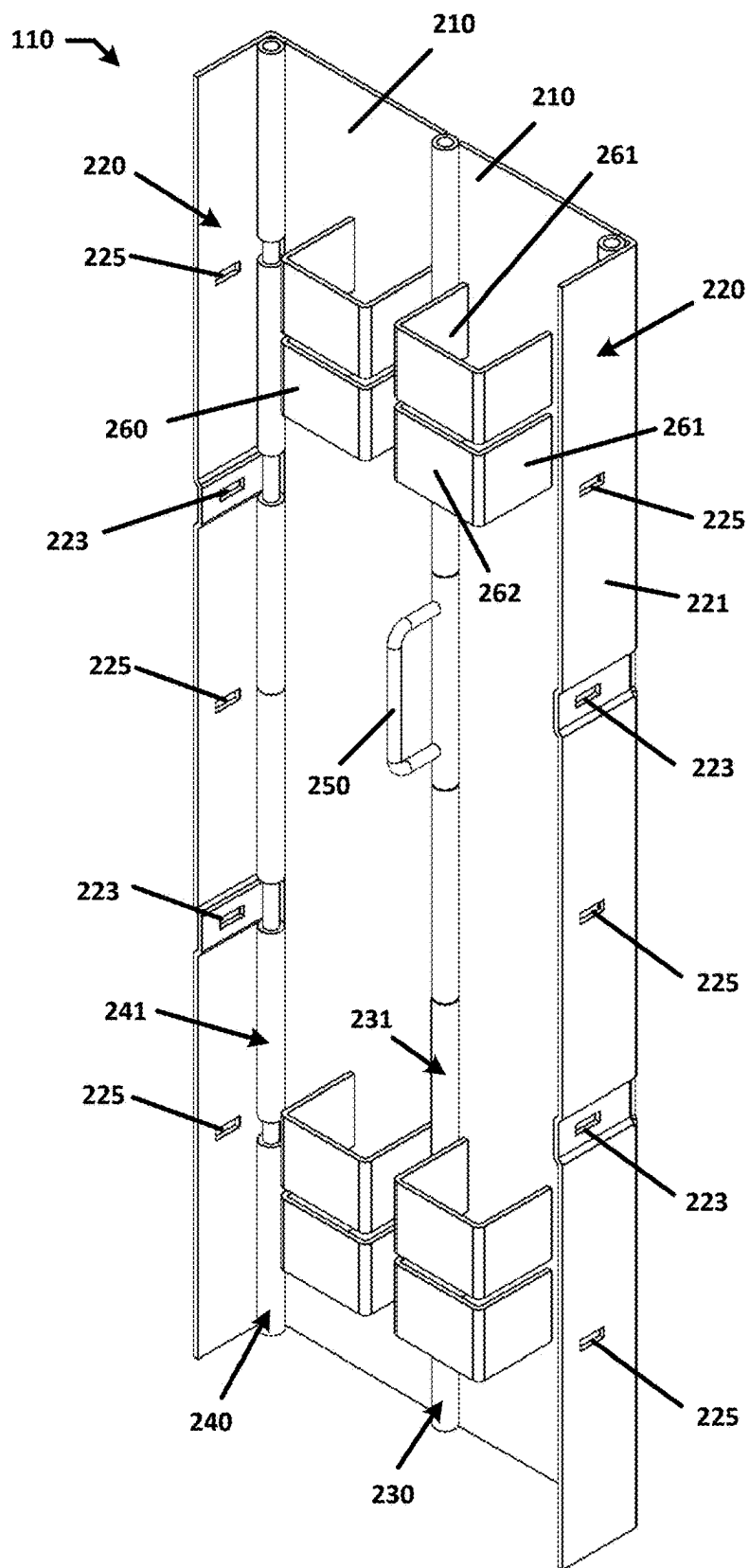


FIG. 2

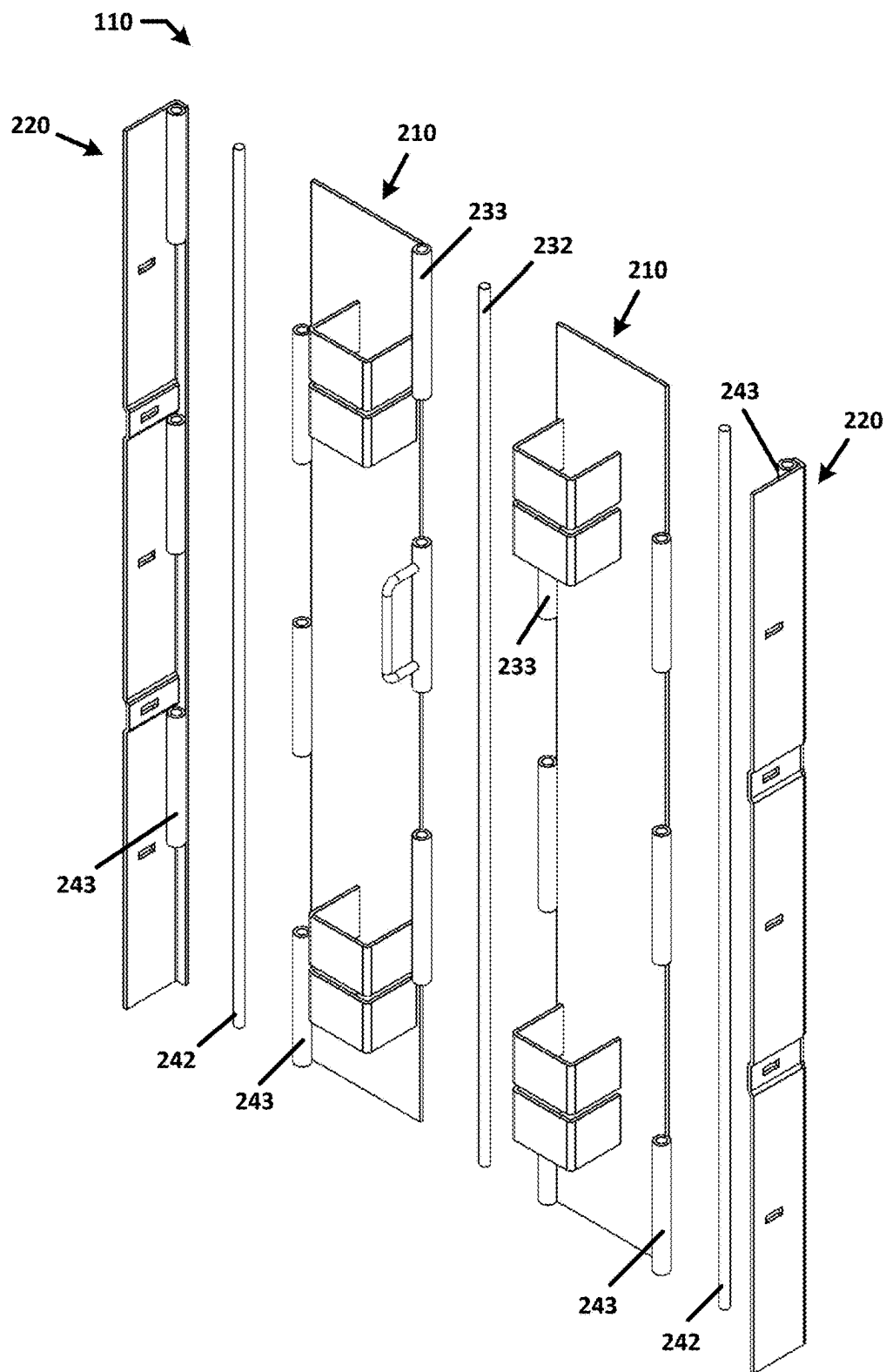


FIG. 3

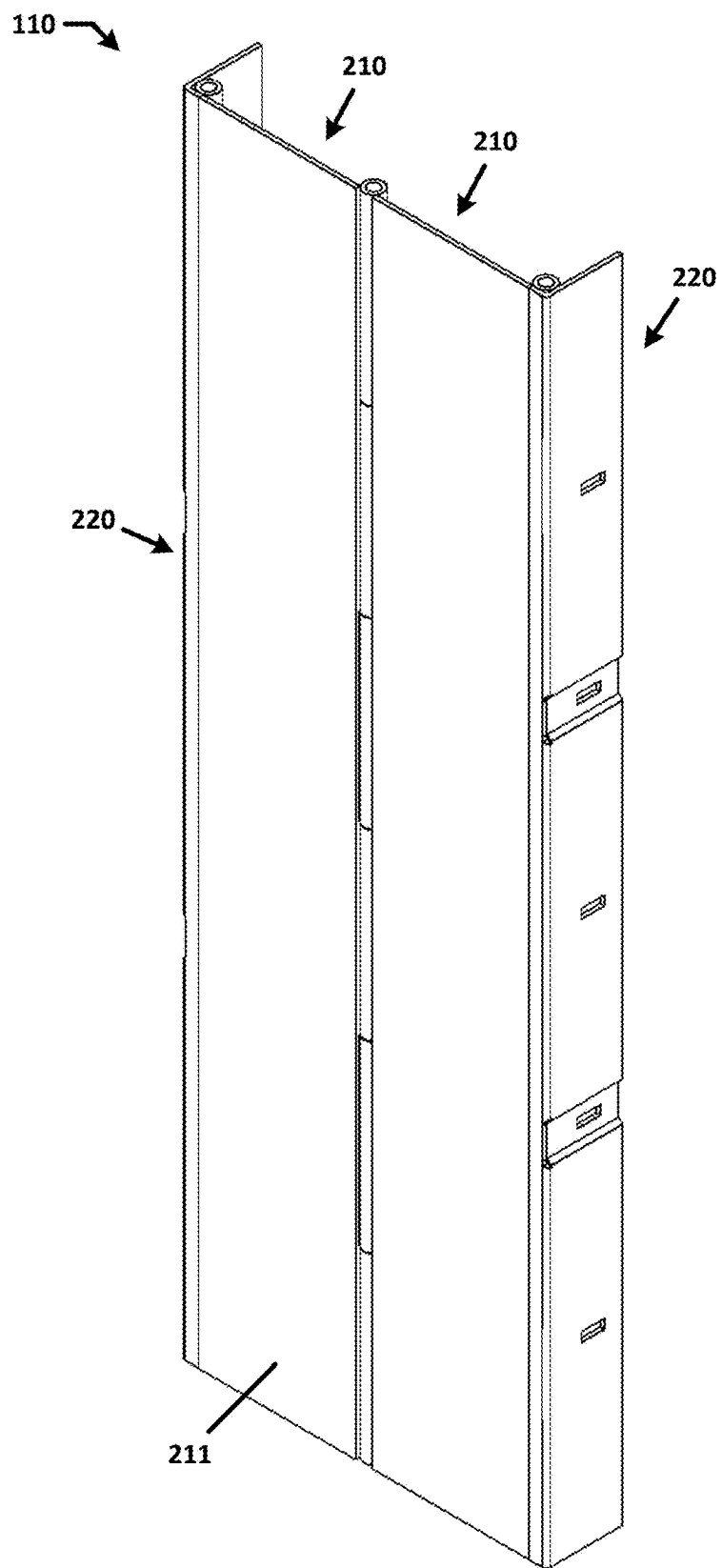


FIG. 4

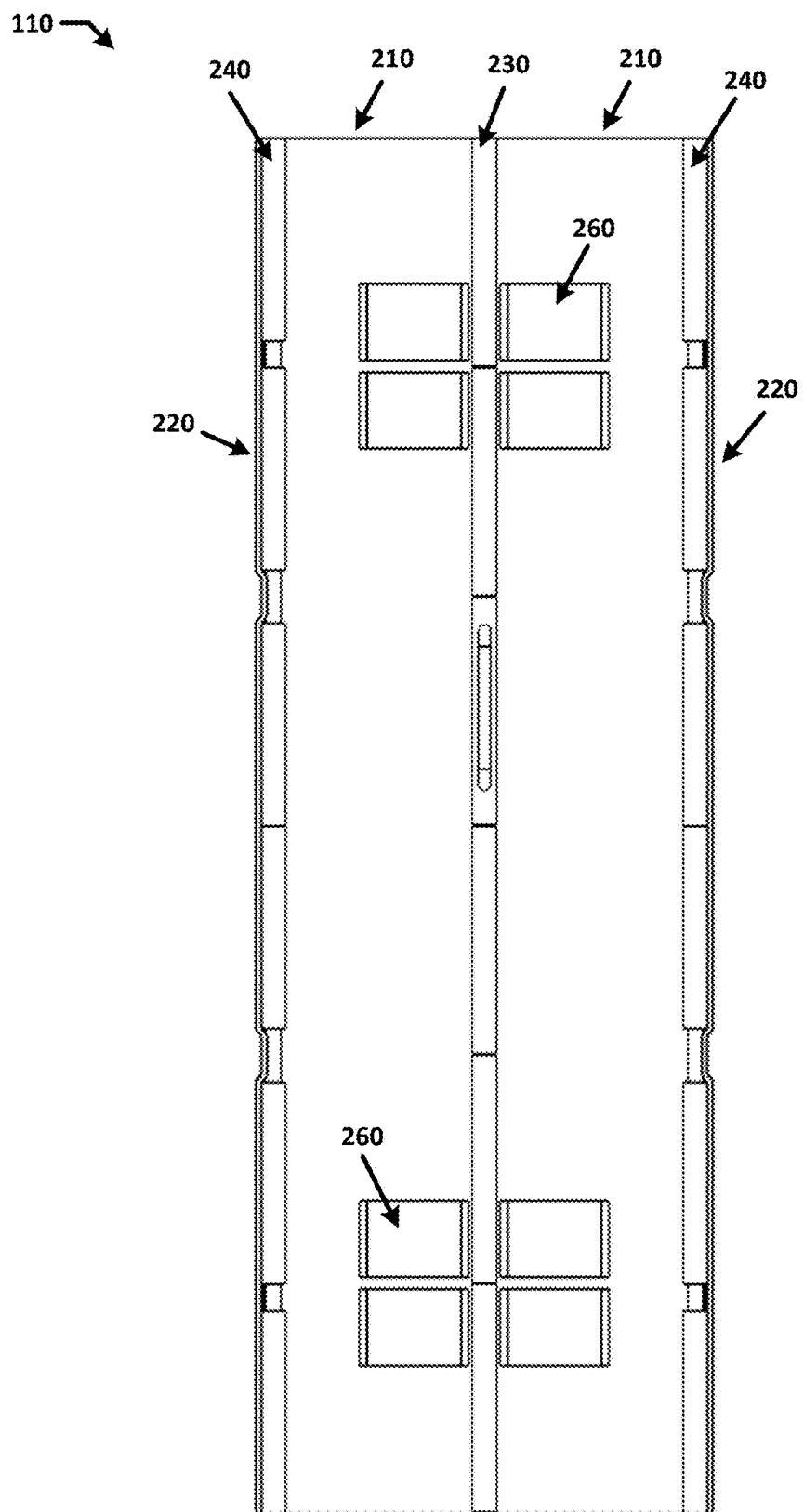


FIG. 5

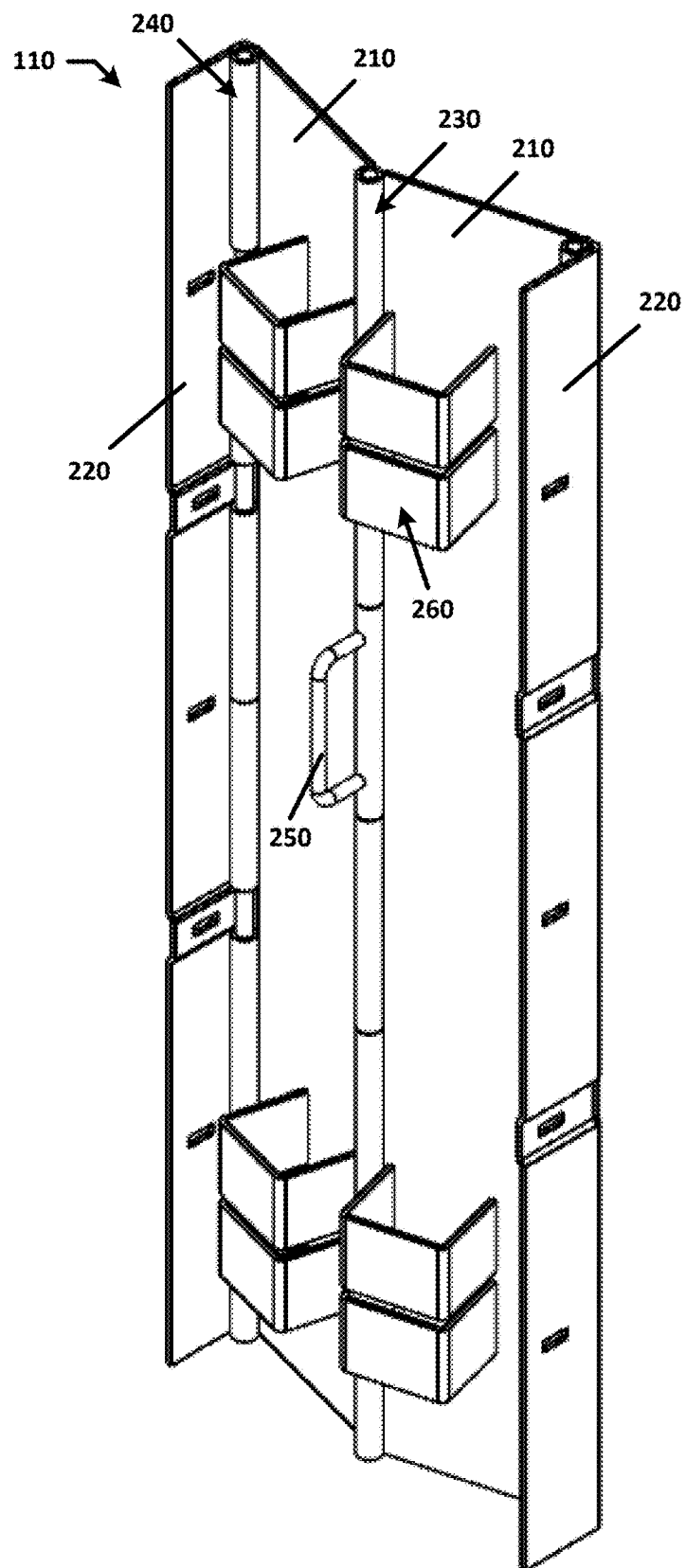


FIG. 6

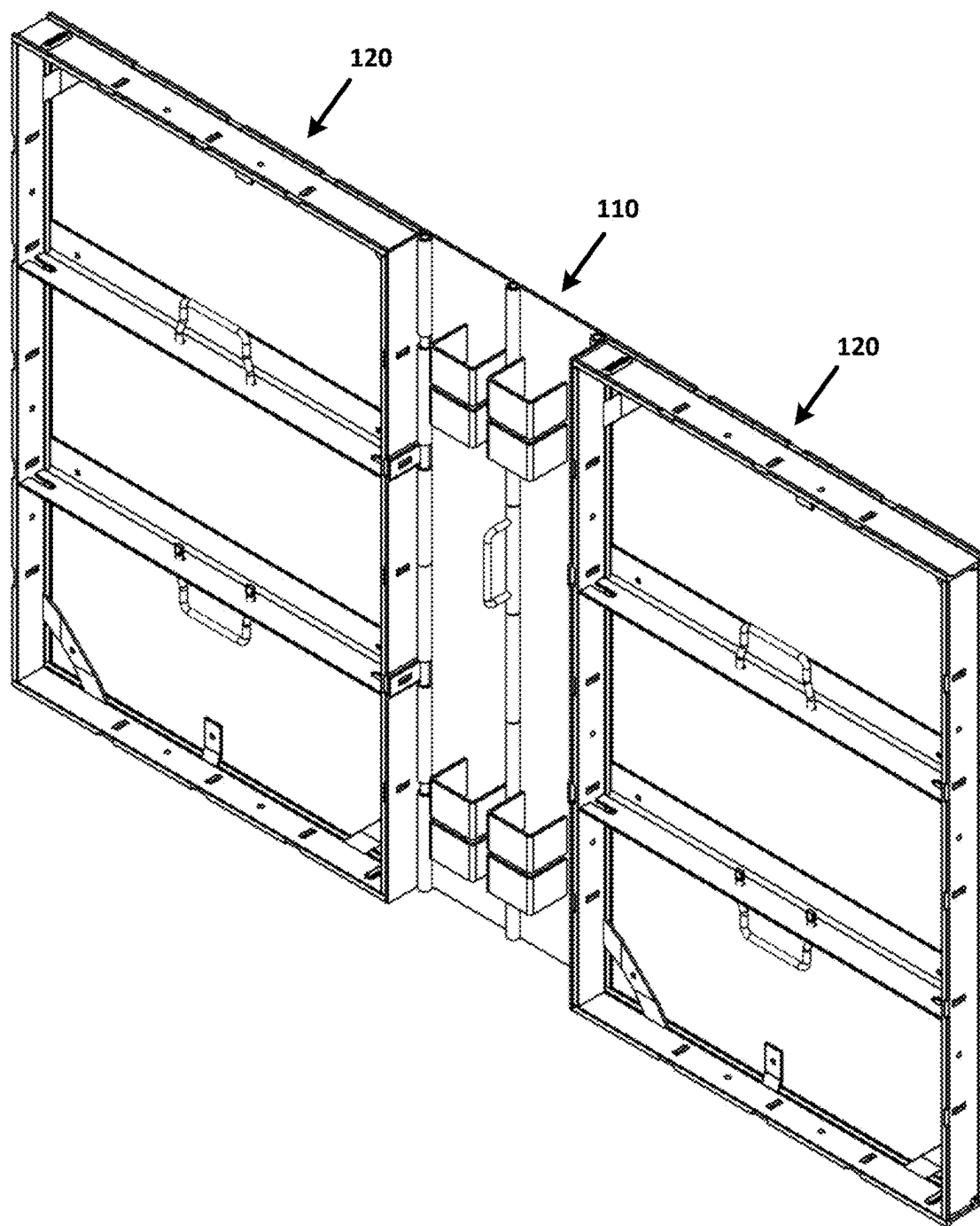


FIG. 7

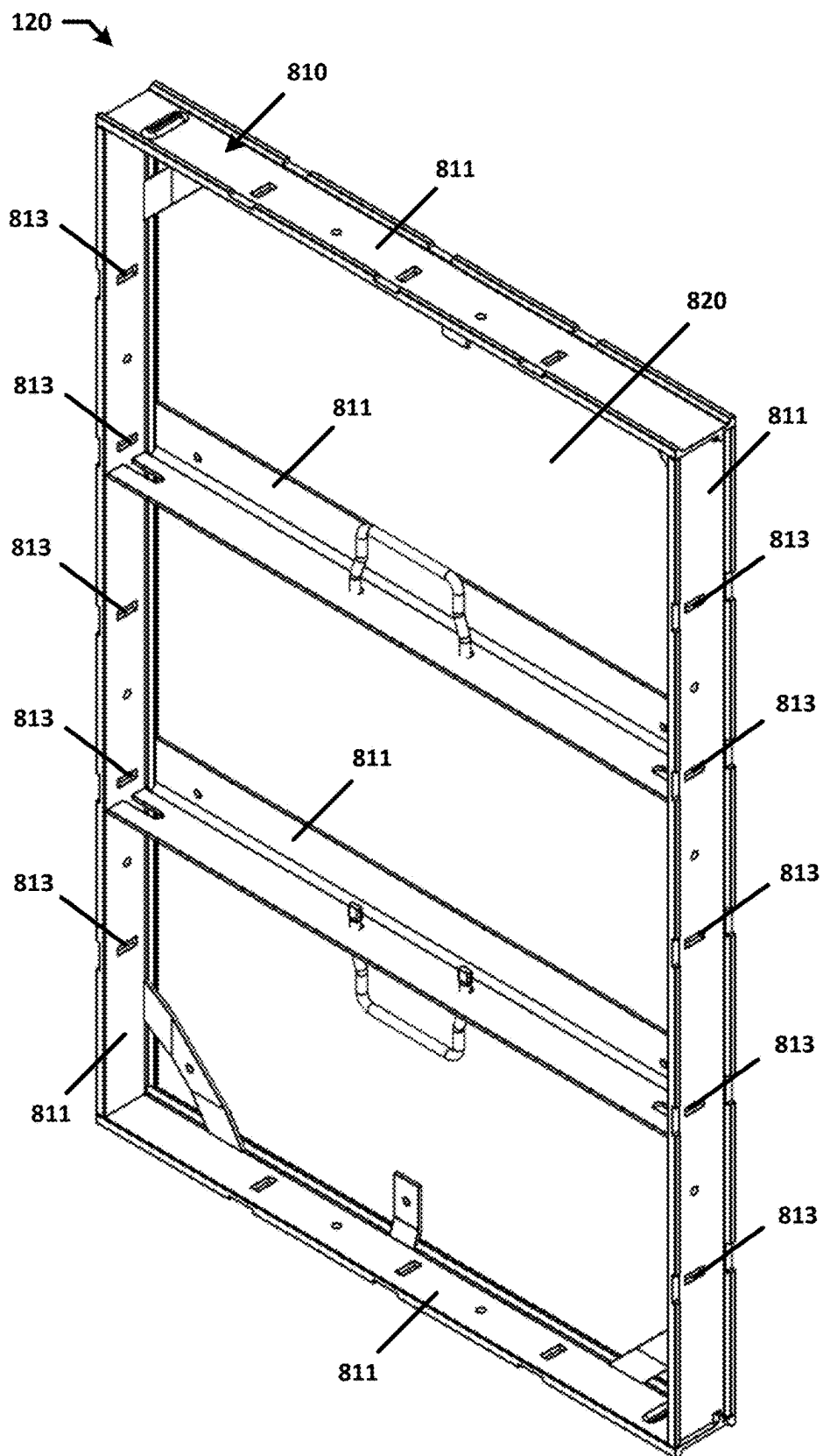


FIG. 8

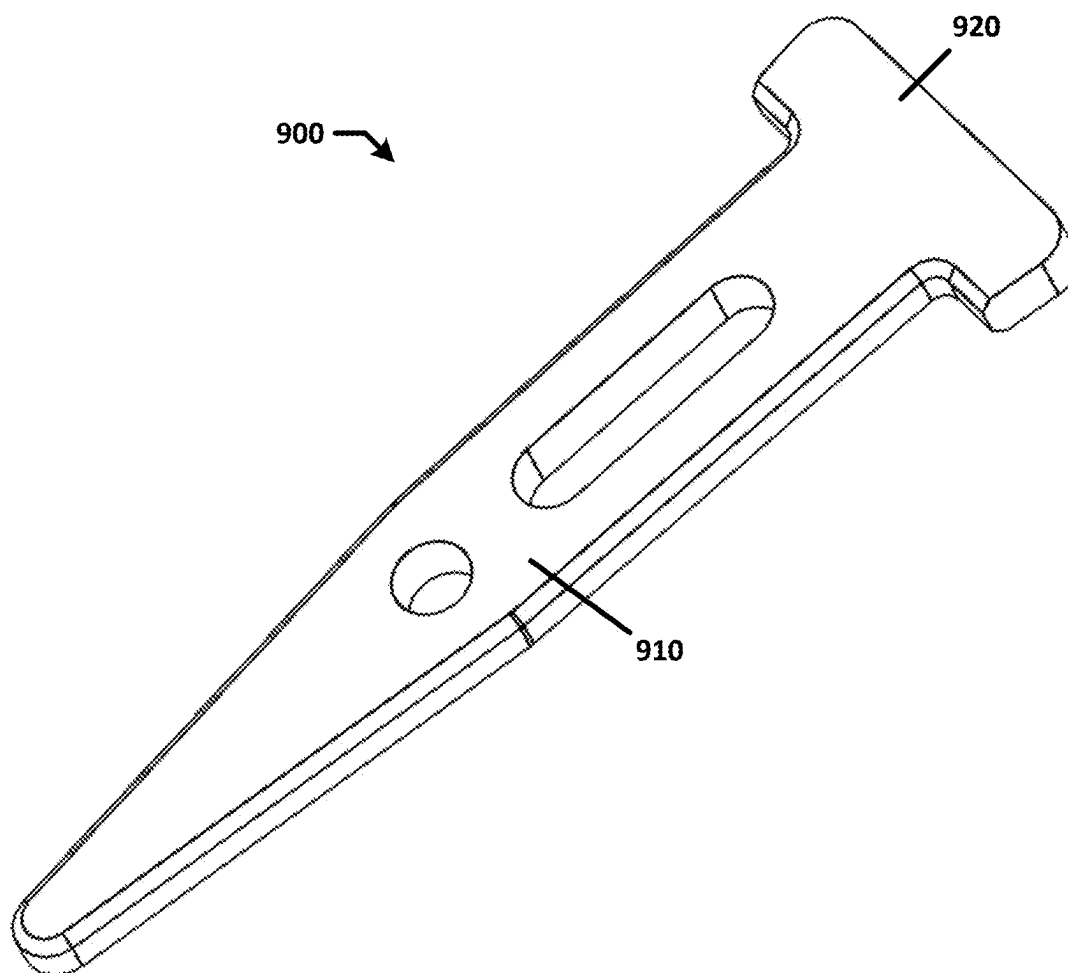


FIG. 9

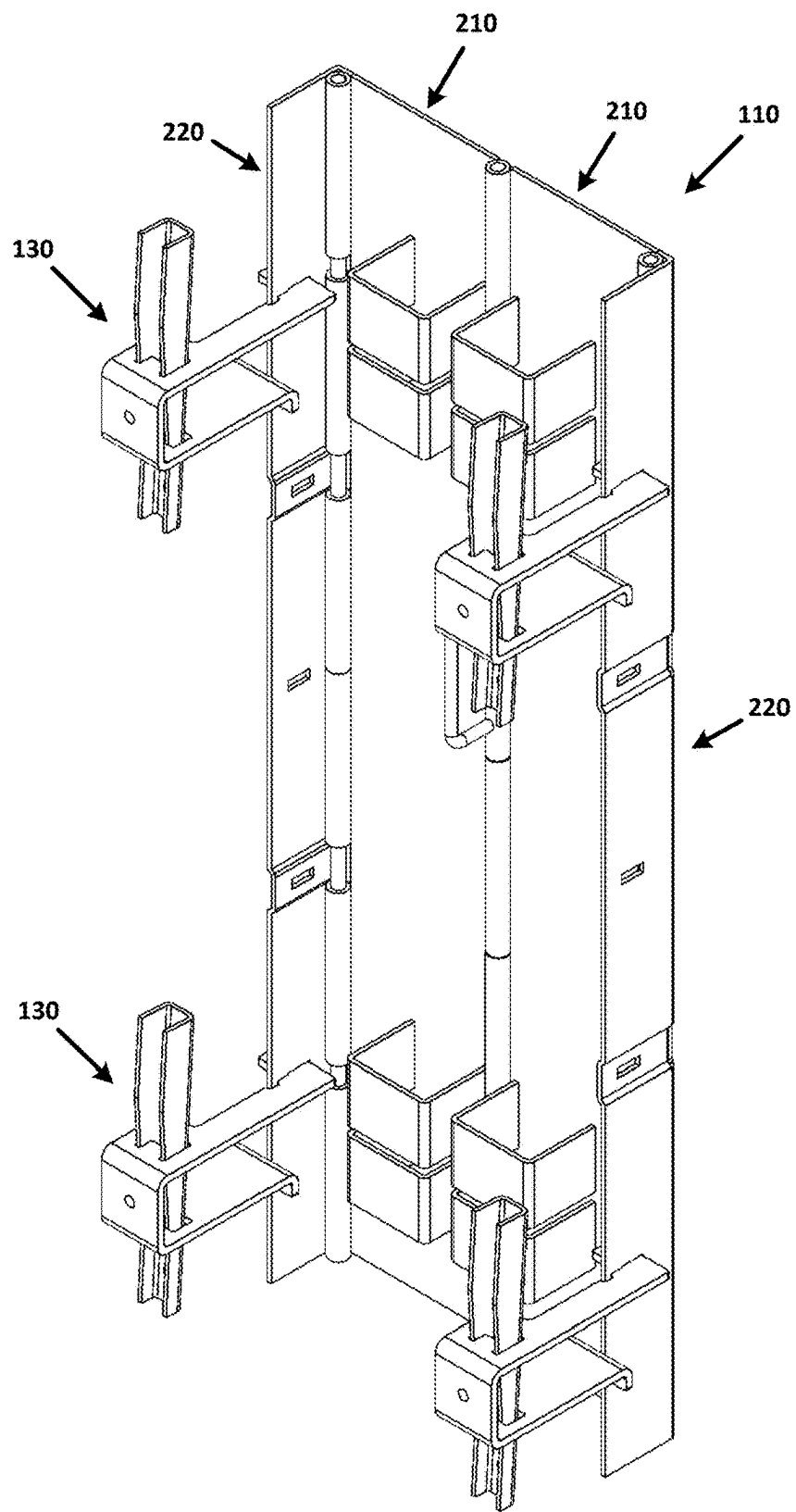


FIG. 10

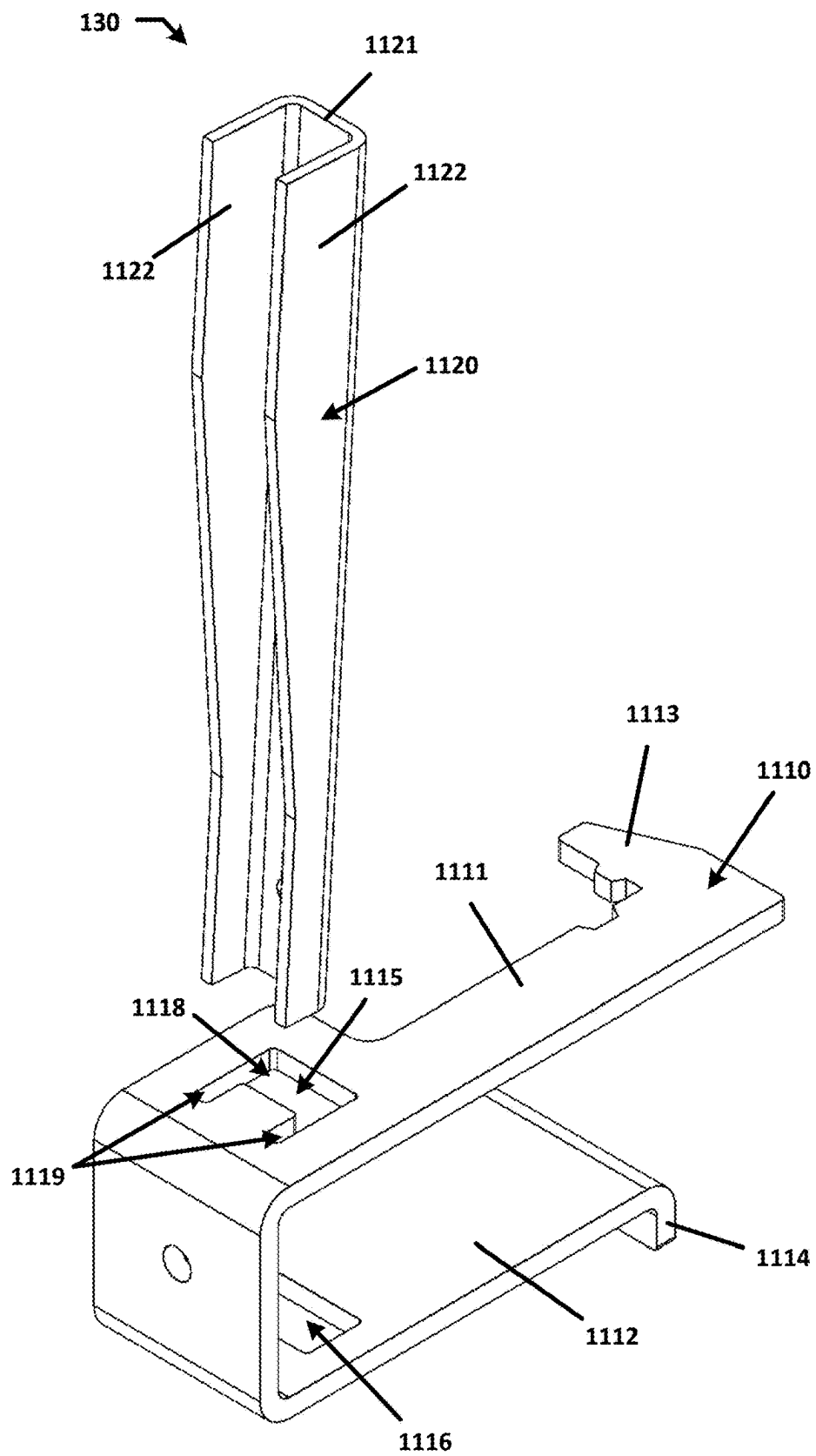


FIG. 11

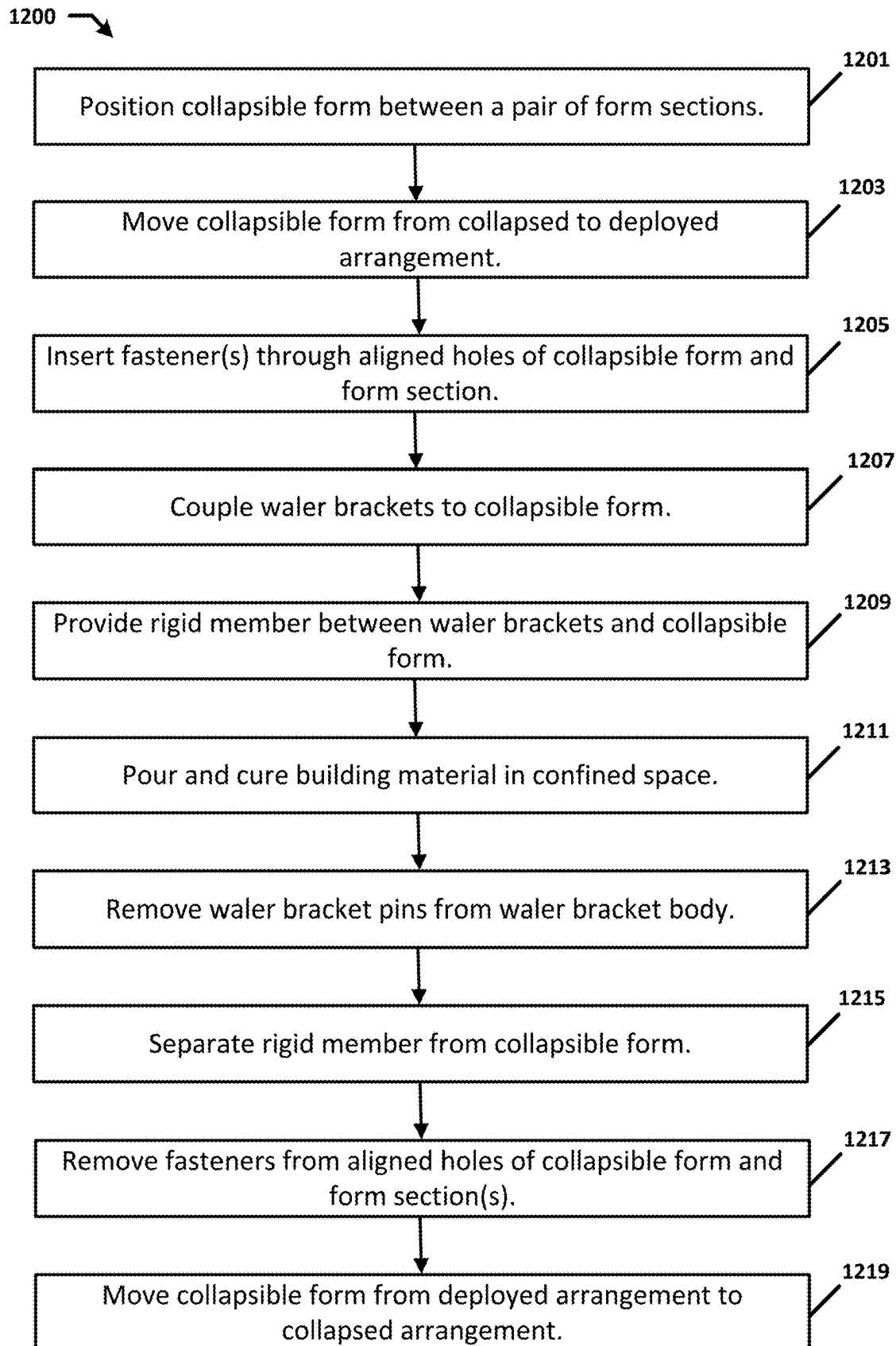


FIG. 12

COLLAPSIBLE FORM

[0001] This application claims the benefit of U.S. provisional application entitled “COLLAPSIBLE FORM” filed Mar. 1, 2024, and assigned Ser. No. 63/560,606, the entire disclosure of which is hereby expressly incorporated by reference.

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

[0002] The disclosure generally relates to concrete casting.

Brief Description of Related Technology

[0003] Generally, one or more forms (e.g., steel plywood form systems, aluminum form systems) are used to define a confined space in which concrete is poured to form a concrete wall or casting. After a concrete wall or casting has been poured and has cured to a suitable strength, the form(s) may be stripped (e.g., removed, separated) from the wall or casting. When stripping the forms from the concrete wall or casting, it may be difficult to remove the first form. Specifically, pressure created by the weight of the concrete may cause the forms to become wedged tightly together, making it difficult to remove the first form. Once the first form is stripped or removed, it may be easier to remove subsequent forms from the wall or casting.

SUMMARY OF THE DISCLOSURE

[0004] In accordance with one aspect of the disclosure, a collapsible concrete form includes a pair of inner panels hingedly coupled to one another and a pair of outer flanges configured to abut a pair of adjacent form sections, such that the pair of outer flanges and the pair of inner panels are disposed between the pair of adjacent form sections. A first one of the pair of outer flanges is hingedly coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges is hingedly coupled to a second one of the pair of inner panels. The collapsible concrete form further includes a handle protruding from one of the pair of inner panels or a hinge coupling the pair of inner panels. The handle is operable to move the pair of inner panels and the pair of outer flanges between a deployed arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner panels are askew.

[0005] In accordance with another aspect of the disclosure, a concrete forming system includes a collapsible form having a pair of inner panels hingedly coupled to one another, a pair of outer flanges configured to abut an adjacent form section. A first one of the pair of outer flanges is hingedly coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges is hingedly coupled to a second one of the pair of inner panels. The collapsible form further includes a pair of waler bracket mounting holes configured to receive a waler bracket. A first one of the pair of waler bracket mounting holes extends through the first one of the pair of outer flanges and a second one of the pair of waler bracket mounting holes extends through the second one of the pair of outer flanges. The concrete forming system further includes a pair of waler brackets configured to be inserted into the pair of waler bracket mounting holes. The pair of inner panels and the pair of outer flanges are configured to move between a deployed arrangement in

which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner panels are askew.

[0006] In accordance with another aspect of the disclosure, a collapsible concrete form includes an inner portion including a pair of inner panels pivotably coupled to one another. The inner portion is configured to move between a linear arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner walls are askew. The collapsible form also includes a pair of outer flanges configured to abut a pair of adjacent form sections, such that the pair of outer flanges and the inner portion are disposed between the pair of adjacent form sections. A first one of the pair of outer flanges is pivotably coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges is pivotably coupled to a second one of the pair of inner panels.

[0007] In connection with any one of the aforementioned aspects, the devices and systems described herein may alternatively or additionally include any combination of one or more of the following aspects or features. A locking brace may protrude from one of the pair of inner panels. The locking brace may include a pair of projections extending from one of the pair of inner panels and a bearing panel extending between the pair of projections and disposed parallel to one of the pair of inner panels. The collapsible form may include a pair of waler bracket mounting holes configured to receive a waler bracket. A first one of the pair of waler bracket mounting holes may extend through the first one of the pair of outer flanges. A second one of the pair of waler bracket mounting holes may extend through the second one of the pair of outer flanges. The pair of waler bracket mounting holes may be configured to receive a pair of waler brackets. In the deployed arrangement, the pair of waler brackets may be configured to support a rigid member disposed between the pair of waler brackets and the locking brace. The pair of waler brackets and the locking brace are configured to abut the rigid member, preventing the pair of inner panels and the pair of outer flanges from moving from the deployed arrangement. The collapsible form may include a pair of side hinges. A first one of the pair of outer flanges and a first one of the pair of inner panels may collectively form a first one of the pair of side hinges. A second one of the pair of outer flanges and a second one of the pair of inner panels may collectively form a second one of the pair of side hinges. The pair of inner panels may collectively form the hinge coupling the pair of inner panels. In the deployed arrangement, the pair of inner panels are configured to define a planar boundary of confined space in which concrete is poured. A form coupling hole may extend through one of the pair of outer flanges. The form coupling hole may be configured to receive a fastener for coupling a respective outer flange of the pair of outer flanges to an adjacent form section of the pair of adjacent form sections. The bearing panel may be configured to abut a rigid member disposed between the pair of waler brackets and the locking brace. When the rigid member is disposed between the pair of waler brackets and the locking brace, the pair of outer flanges may be configured to abut the rigid member. A handle may protrude from one of the pair of inner panels or a hinge coupling the pair of inner panels. The handle may be operable to move the pair of inner panels and the pair of outer flanges between the deployed arrangement and the collapsed arrangement. A form coupling hole may extend through one of the pair of outer flanges. A collapsible form

system may include a form section having a form section panel, a frame coupled to the form section panel and disposed along a perimeter of the form section panel, and a form section coupling hole extending through the frame. A fastener may be configured to be inserted through the form coupling hole and the form section coupling hole, coupling the collapsible form and the form section. When the collapsible form and the form section are coupled, the pair of inner walls and the form section panel are collinear and collectively define planar boundary of a confined space in which concrete is poured. The form section coupling hole may include a pair of form section coupling holes. The fastener may be configured to be inserted through the form coupling hole and one of the pair of form section coupling holes, coupling the collapsible form and the form section. One of the pair of waler brackets may be configured to be inserted through one of the pair of waler bracket mounting holes and an other of the pair of form section coupling holes.

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0008] For a more complete understanding of the disclosure, reference should be made to the following detailed description and accompanying drawing figures, in which like reference numerals identify like elements in the figures.

[0009] FIG. 1 illustrates a forming system in accordance with one example of the present disclosure.

[0010] FIG. 2 illustrates a front perspective view of a collapsible form in a deployed arrangement in accordance with one example of the present disclosure.

[0011] FIG. 3 illustrates an exploded view of the collapsible form of FIG. 2 in accordance with one example of the present disclosure.

[0012] FIG. 4 illustrates a back perspective view of the collapsible form of FIG. 2 in accordance with one example of the present disclosure.

[0013] FIG. 5 illustrates a back view of the collapsible form of FIG. 2 in accordance with one example of the present disclosure.

[0014] FIG. 6 illustrates the collapsible form of FIG. 2 in a collapsed arrangement in accordance with one example of the present disclosure.

[0015] FIG. 7 illustrates a collapsible form and a pair of adjacent form sections in accordance with one example of the present disclosure.

[0016] FIG. 8 illustrates a form section in accordance with one example of the present disclosure.

[0017] FIG. 9 illustrates a fastener in accordance with one example of the present disclosure.

[0018] FIG. 10 illustrates a plurality of waler brackets coupled to a collapsible form in accordance with one example of the present disclosure.

[0019] FIG. 11 illustrates a waler bracket in accordance with one example of the present disclosure.

[0020] FIG. 12 illustrates a flow chart for using a collapsible form in accordance with one example of the present disclosure.

[0021] While the disclosed devices and systems are susceptible of embodiments in various forms, there are illustrated in the drawing (and will hereafter be described) specific embodiments of the invention, with the understanding that the disclosure is intended to be illustrative, and is not intended to limit the invention to the specific embodiments described and illustrated herein.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0022] As described above, when a building material, for example, concrete is poured into a confined space defined by a plurality of form sections, the weight of the building material may apply pressure to the form sections. A plurality of fasteners (e.g., pins) may couple the form sections to one another. The pressure of the building material may tightly press the form sections and/or fasteners together. After the building material has at least partially cured, the form sections and fasteners may be removed from the building material (e.g., structure formed by the building material). However, because the form sections and the fasteners have been and/or are tightly pressed together under the pressure (e.g., weight) of the building material, it may be difficult to separate the form sections and/or fasteners from one another and/or from the building material (e.g., a structure formed by the building material). In some examples, separation or removal of a first form section may be most difficult, as there is no space or clearance between the plurality of form sections. After removal of a first form section, an opening or gap may be present between the remaining form sections making removal or separation of subsequent form sections easier.

[0023] A collapsible filler or form is described. The collapsible form may be disposed between two or more form sections. The collapsible form and a plurality of form sections may define a confined space in which a building material is poured and/or at least partially cured. The collapsible form may be configured to move between two or more arrangements. For example, the collapsible form may be configured to move between a deployed arrangement in which the collapsible form defines (e.g., at least a portion of) a boundary of the confined space and a collapsed arrangement. In the collapsed arrangement, the collapsible form may have a smaller or more narrow width than in the deployed arrangement. According to the present disclosure, after pouring and/or curing of a building material (e.g., in the confined space), the collapsible form may transition or move from the deployed arrangement to the collapsed arrangement, such that the collapsible form may be more easily separated from the one or more form sections (e.g., due to the additional space created between the collapsible form and the form section when the collapsible form is moved from the deployed arrangement to the collapsed arrangement).

[0024] In accordance with some examples, in the deployed arrangement, the collapsible form has a “U” shape including a pair of inner panels, a pair of outer flanges, and three hinges. In accordance with some examples, in the deployed arrangement, the inner panels may form a web of the “U” shape with an outer flange extending from each end of the pair of inner panels. A hinge (e.g., side hinge) may be disposed between each of the outer flanges and an adjacent inner panel. A hinge (e.g., center hinge) may be disposed between the pair of inner panels. The collapsible form may be included in and/or used with a form system (e.g., steel plywood form systems, aluminum form system) to define a confined space in which a concrete wall or casting may be poured. Specifically, the collapsible form may be attached or coupled to one or more steel plywood form sections or aluminum form sections. Specifically, the collapsible form may be disposed between and coupled to two form sections. The collapsible form may be coupled to the one or more

form sections by one or more fasteners (e.g., wedge bolts, round pins) extending through the collapsible form and an adjacent form section.

[0025] The pair of inner panels and the pair of side flanges may be hingedly or pivotably coupled to one another, such that the collapsible form may move between a deployed arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner panels are askew. In the deployed arrangement, the pair of inner panels may define a planar boundary of a confined space in which concrete may be poured to form a wall or casting. In some examples, the collapsible form may include a handle operable to move the collapsible form between the deployed and collapsed arrangements. When the collapsed form is in the deployed arrangement a rigid member, for example, a 2"x4" piece of lumber may be placed between a pair of waler brackets coupled to the collapsible form and a locking brace of the collapsible form. The rigid member may abut the waler brackets and the locking brace, preventing the collapsible form (e.g., pair of inner panels) from moving from the deployed arrangement in which the pair of inner panels are collinear while concrete is poured in a confined space defined by the pair of inner panels. The collapsible form may be configured to withstand the hydraulic pressure of concrete poured in a confined space defined by the collapsible form.

[0026] After the concrete has cured, the rigid member, waler brackets, and fasteners coupling the collapsible form to adjacent form sections are removed, and the collapsible form may be moved (e.g., collapsed) from deployed arrangement to the collapsed arrangement. For example, a user may pull on the handle, causing the collapsible form to move from the deployed arrangement to the collapsed arrangement. The collapsible form may have a different (e.g., narrower) shape in the collapsed arrangement than in the deployed arrangement, allowing the collapsible form to be easily removed from a concrete wall or casting. This collapsible form can be used while pouring basements, many different precast castings, above grade walls, or in any casting where pressure from the concrete makes it difficult to remove a first form. The collapsible form can be used with steel plywood forms, aluminum forms, or any other type of form.

[0027] Referring to FIG. 1, a forming system 100 is illustrated in accordance with one example of the present disclosure. As shown in FIG. 1, the form system 100 may include a collapsible form 110, two or more form sections 120, two or more waler brackets 130, and at least one rigid member 140. In accordance with some examples, the forming system 100 may be a concrete forming system. According to other examples, the forming system 100 may be used with another type of building material, for example, plaster, asphalt, or the like. According to the present disclosure, the collapsible form 110 may be removably coupled to two or more form sections 120. For example, as shown in FIG. 1, the collapsible form 110 may be directly coupled to and disposed between a pair of adjacent form sections 120.

[0028] Referring to FIG. 2, a back perspective view of the collapsible form 110 is illustrated in accordance with one example of the present disclosure. The collapsible form 110 may include two or more inner panels 210 and two outer flanges 220. According to some examples, as shown in FIG. 1, the collapsible form 110 may include a pair of (e.g., two) inner panels 210. However, the present disclosure is not

limited thereto and in other examples, another number of inner panels 210 may be included. For example, three, four, or more than four inner panels 210 may be included.

[0029] As noted above, the collapsible form 110 may be configured to move between a deployed arrangement in which the inner panels 210 are collinear and a collapsed arrangement in which the inner panels 210 are askew (e.g., disposed at a non-zero angle with respect to one another). As shown in FIG. 2, the collapsible form 110 is disposed in a deployed arrangement.

[0030] In the deployed arrangement, the inner panels 210 may define (e.g., at least a portion of) a (e.g., planar) boundary of a confined space in which a building material, for example, concrete, asphalt, plaster, or the like, may be poured. Additionally, the building material may be at least partially cured within the confined space, a boundary of which is at least partially defined by the inner panels 210 when the collapsible form 110. For example, the building material may be cured within the confined space at least until the building material is self-supporting. The confined space may correspond, for example, in size and/or shape, to a structure to be formed by the building material.

[0031] In the collapsed arrangement, a width of the collapsible form 110, for example, a distance between the outer flanges 220 of the collapsible form 110 may be smaller than a width of the of the collapsible form 110 in the deployed arrangement. Accordingly, as noted above, after a building material has been poured into and at least partially cured within the confined space, collapsible form 110 may be moved (e.g., collapsed) from the deployed arrangement to the collapsed arrangement before or as the collapsible form 110 is separated from and/or removed from between the pair of adjacent form sections. As described above, moving the collapsible form from the deployed arrangement to the collapsed arrangement may facilitate or improve the ease with which the collapsible form 110 may be separated from and/or removed from between the pair of adjacent form sections.

[0032] Similarly, in some examples, the collapsible form 110 may be inserted between the pair of adjacent form sections, for example, before pouring of the building material, in the collapsed arrangement and moved (e.g., expanded) to the deployed arrangement while already disposed between the pair of adjacent form sections.

[0033] Still referring to FIG. 2, the inner panels 210 may be movably coupled to one another. For example, the inner panels 210 may be hingedly or pivotably coupled to one another, such that the inner panels 210 are free to move relative to one another. As described hereinafter in greater detail, the inner panels 210 may be movably coupled to one another so as to be collinear when the collapsible form 110 is in a deployed arrangement and askew when the collapsible form 110 is in a collapsed arrangement. One or more center hinge(s) 230 may rotatably couple the inner panels 210.

[0034] FIG. 3 illustrates an exploded view of the collapsible form 110 of FIG. 2. Referring to FIGS. 2 and 3, the center hinge 230 may include a barrel 231 and a pin 232. The barrel 231 may be comprised of a plurality of barrel segments 233. The barrel 231 may be comprised of one or more barrel segments 233 fixedly attached to a first one of the pair of inner panels 210 and one or more barrel segments 233 fixedly attached to a second one of the pair of inner panels 210. According to some examples, the one or more barrel

segments **233** fixedly attached to the first one of the pair of inner panels **210** and the one or more barrel segments **233** fixedly attached to the second one of the pair of inner panels **210** may be integrally formed with their respective inner panel **210**. According to other examples, one or more barrel segments **233** may be fixedly attached to a respective one of the pair of inner panels **210** using one or any combination of: one or more fasteners (e.g., nuts, bolts, rivets, screws, or the like), welding, adhesive bonding, or the like.

[0035] The one or more barrel segments **233** fixedly attached to the first one of the pair of inner panels **210** and the one or more barrel segments **233** fixedly attached to the second one of the pair of inner panels **210** may be moved or positioned such that a hole or channel extending through each of the barrel segments **233** is aligned. The pin **232** may be inserted or disposed within the channel of the plurality of barrel segments **233**, movably coupling the pair inner panels **210** to one another. For example, the pair of inner panels **210** may be coupled such that they may rotate or pivot relative to one another about the pin **232**.

[0036] FIG. 4 illustrates a front perspective view of the collapsible form **110** of FIG. 2 in a deployed state. The inner panels **210** may be configured to define (e.g., at least a portion of) a boundary of a confined space in which concrete is poured. For example, when the collapsible form **110** is in the deployed state, as shown in FIGS. 2 and 4, the inner panels **210** may be configured to define (e.g., at least a portion of) a boundary of a confined space in which concrete is poured. A front surface or face **211** of the inner panels **210** may be configured to define (e.g., at least a portion of) a boundary of a confined space in which concrete is poured. According to some examples, the face **211** of the inner panels **210** may be planar, such that the portion of a boundary of a confined space in which concrete may be poured defined by each of the inner panels **210** is planar. According to the present disclosure, the inner panels **210** of the collapsible form **110** may be collinear when the collapsible form **110** is in the deployed arrangement, such that the inner panels **210** collectively define a planar boundary of the confined space.

[0037] Returning to FIG. 2, the collapsible form **110** may further include one or more locking braces **260**. As shown in FIG. 1, when the collapsible form **110** is disposed between a pair of adjacent form sections **120**, two or more waler brackets **130** may be coupled to the collapsible form **110**. The waler brackets **130** may be (e.g., removably) coupled to the collapsible form **110** in pairs. Each pair of waler brackets **130** may be configured to support a rigid member **140** disposed between the pair of waler brackets **130** and one or more locking braces **260** extending from the collapsible form **110**. The one or more locking braces **260** may be configured to abut a rigid member **140** disposed between a pair of waler brackets **130** and the collapsible form **110**. The one or more locking braces **260** may exert a force or bear on the rigid member **140** (e.g., supported by the pair of waler brackets **130** and disposed between the pair of waler brackets **130** and the collapsible form **110**), preventing the collapsible form **110** from moving from the deployed arrangement (e.g., to the collapsed arrangement).

[0038] According to the present disclosure, when a building material is poured in a confined space having a boundary at least partially defined by the inner panels **210** of the collapsible form **110**, pressure created by a weight of the building material may exert an (e.g., outward) force on the

inner panels **210**, biasing the collapsible form **110** to move from the deployed arrangement, in which the inner panels **210** are collinear, to the collapsed arrangement, in which the inner panels **210** are askew. However, when the rigid member **140** is disposed between a pair of waler brackets **130** and the one or more locking braces **260**, the rigid member **140** may exert a force on the one or more locking braces **260** that is equal to and opposite the outward force exerted (e.g., due to the pressure of the building material in the confined space) on the inner panels **210**, preventing the collapsible form **110** from moving from the deployed arrangement.

[0039] Returning to FIG. 2, each of the one or more locking braces **260** may extend from a respective one of the inner panels **210**. Each of the one or more locking braces **260** may include at least one projection **261** extending from one of the pair of inner panels **210** and a bearing surface or bearing panel **262** configured to engage (e.g., abut, contact) a rigid member disposed between a pair of waler brackets and the collapsible form **110**. According to some examples, as shown in FIG. 2, each locking brace **260** may include two projections **261** extending from one of the inner panels **210**. According to some examples, as shown in FIG. 2, the bearing panel **262** may extend between the pair of projections **261**. However, the present disclosure is not limited thereto, and in other examples, the bearing panel **262** may be coupled to a single projection **261** included in the locking brace **260**. According to some examples, the bearing panel **262** may be disposed parallel to an inner panel **210** to from which the locking brace **260** extends.

[0040] Still referring to FIG. 2, the collapsible form **110** may include a pair of outer flanges **220**. The pair of outer flanges **220** may be configured to engage a pair of adjacent form sections, for example, the adjacent form sections **120** illustrated in FIG. 1, when the collapsible form **110** is in the deployed arrangement. Each of the outer flanges **220** may engage a respective one of the pair of adjacent form sections, such that the collapsible form **110** is disposed between the pair of adjacent form sections. For example, each of the pair of outer flanges **220** may be configured to abut or contact a respective adjacent form section. In accordance with some examples, each outer flange **220** may include a contact surface **221** configured to interface with engage (e.g., abut, contact) a form section adjacent to the respective outer flange **220**.

[0041] As described above, the collapsible form **110** may be configured to move between the deployed arrangement (as illustrated in FIG. 2) and a collapsed arrangement. According to the present disclosure, each of the pair of outer flanges **220** may be movably coupled to a respective one of the inner panels **210**, facilitating movement of the collapsible form **110** between the deployed and collapsed arrangements. For example, a first one of the pair of outer flanges **220** may be movably coupled to a first one of the inner panels **210** and a second one of the pair of outer flanges **220** may be movably coupled to a second one of the inner panels **210**. For example, the outer flanges **220** may be hingedly or pivotably coupled to a respective one of the inner panels **210**, such that the outer flange **220** is free to move relative to a respective one of the inner panels **210**.

[0042] According to some examples, when the collapsible form **110** is in the deployed arrangement, as illustrated in FIG. 2, the pair of outer flanges **220** may be disposed parallel to one another.

[0043] Referring to FIGS. 2 and 3, according to some examples, the collapsible form 110 may include a pair of side hinges 240 movably coupling the outer flanges 220 to their respective inner panels 210.

[0044] Each side hinge 240 may include a barrel 241 and a pin 242. The barrel 241 may be comprised of a plurality of barrel segments 243. The barrel 241 may be comprised of one or more barrel segments 243 fixedly attached to a respective one of the outer flanges 220 and one or more barrel segments 243 fixedly attached to a respective one of the inner panels 210. According to some examples, the one or more barrel segments 243 fixedly attached to the outer flange 220 and the one or more barrel segments 243 fixedly attached to the inner panel 210 may be integrally formed with the outer flange 220 and/or inner panel 210, respectively. According to other examples, the one or more barrel segments 243 may be fixedly attached to the outer flange 220 and/or the one or more barrel segments 243 may fixedly attached to the inner panel 210 using one or any combination of: one or more fasteners (e.g., nuts, bolts, rivets, screws, or the like), welding, adhesive bonding, or the like.

[0045] The one or more barrel segments 243 fixedly attached to the outer flange 220 and the one or more barrel segments fixedly attached to the inner panel 210 may be aligned. For example, the one or more barrel segments 243 fixedly attached to the outer flange 220 and the one or more barrel segments fixedly attached to the inner panel 210 may be aligned such that a hole or channel extending through each of the barrel segments 243 is aligned. The pin 242 may be inserted or disposed within the channel of the plurality of barrel segments 243, movably coupling the outer flange 220 and the inner panel 210. For example, the outer flange 220 and the inner panel 210 may be coupled such that they may rotate or pivot relative to one another about the pin 242.

[0046] FIG. 5 illustrates a back view of the collapsible form 110 of FIG. 1 in accordance with one example of the present disclosure.

[0047] Referring to FIG. 6, the collapsible form 110 of FIG. 2 is illustrated in a collapsed state in accordance with one example of the present disclosure. As noted above, the collapsible form 110 may be configured to move between a deployed arrangement (e.g., illustrated in FIGS. 2, 4, and 5) and a collapsed arrangement (e.g., illustrated in FIG. 6). As described above, the inner panels 210 may be movably coupled to one another and the outer flanges 220 may be movably coupled to a respective one of the inner panels 210. The movable connections or couplings between the inner panels 210 and the outer flanges 220 and a respective one of the inner panels 210 described above may facilitate or allow movement of the collapsible form 110 between the deployed arrangement (e.g., as illustrated in FIGS. 2, 4, and 5) and the collapsed arrangement (e.g., as illustrated in FIG. 6).

[0048] Referring to FIG. 2, in some examples, the collapsible form 110 may further include a handle 250. The handle 250 may be operable to move the collapsible form 110 between the deployed and collapsed arrangements. For example, the handle 250 may be grabbed or held by a person who may exert a force (e.g., push, pull) on the collapsible form 110, via the handle 250, moving the collapsible form 110 between the deployed and collapsed arrangements. According to some examples, a person may grab and hold one of an inner panel 210 or an outer flange 220 (e.g., with a second hand) while a force is exerted on the collapsible form 110 via the handle 250, such that, the collapsible form

110 is moved between the deployed and collapsed arrangements. According to other examples, one or both of the outer flanges 220 may be coupled to an adjacent form section while a force is exerted on the collapsible form 110 via the handle 250, such that, the collapsible form 110 is moved between the deployed and collapsed arrangements.

[0049] According to some examples, as shown in FIG. 2, the handle 250 may be coupled to the center hinge 230. For example, the handle 250 may be coupled to a barrel segment 233 coupled to one of the pair of inner panels 210. According to some examples, the handle 250 may be coupled to one of the inner panels 210, for example, to a back surface of one of the inner panels 210.

[0050] As shown in FIG. 6, in the collapsed arrangement, the inner panels 210 of the collapsible form 110 may be askew or disposed at an angle (e.g., other than zero) with respect to one another. According to the present disclosure, a width of the collapsible form 110 may change as the collapsible form 110 moves between the deployed and collapsed arrangements. The width of the collapsible form 110 may be defined as a distance between the pair of outer flanges 220. For example, the width of the collapsible form 110 may be a distance between the contact surfaces 221 of the pair of outer flanges 220. According to the present disclosure, because the inner panels 210 are askew in the collapsed arrangement, a width of the collapsible form 110 in the collapsed arrangement may be smaller than a width of the collapsible form 110 in the deployed arrangement.

[0051] Referring to FIG. 7, the collapsible form 110 may be coupled to one or more form sections 120. For example, as described above and illustrated in FIG. 7, the collapsible form 110 may be disposed between a pair of adjacent form sections 120. The collapsible form 110 and the one or more form sections 120 may collectively define the confined space in which a building material may be poured and/or (e.g., at least partially) cured. In some examples, the collapsible form 110 may be (e.g., directly) coupled to a pair of adjacent form sections 120. As described above, each of the outer flanges 220 may be configured to engage or interface with an (e.g., directly) adjacent form section 120, such that the collapsible form 110 is disposed between a pair of (e.g., directly) adjacent form sections.

[0052] According to the present disclosure, the collapsible form 110 may be configured to move between the collapsed arrangement (e.g., as shown in FIG. 6) and the deployed arrangement (e.g., as shown in FIGS. 2 and 7), such that the collapsible form 110 may be inserted between and/or removed from between a pair of (e.g., directly) adjacent form sections 120 in the collapsed arrangement and define (e.g., at least a portion of) a boundary of a confined space in the deployed arrangement.

[0053] Referring to FIG. 7, according to the present disclosure, a distance between a pair of adjacent form sections 120, between which the collapsible form 110 is inserted or disposed may be equal to or less than a width of the collapsible form 110 when the collapsible form 110 is in the deployed arrangement. Accordingly, the collapsible form 110 may be inserted between the pair of adjacent form sections, in the collapsed arrangement and as the collapsible form 110 is moved to the deployed arrangement, the pair of outer flanges 220 may be brought into contact with respective ones of the pair of adjacent form sections 120. In accordance with some examples, a distance between the pair of adjacent form sections 120, between which the collapsible form 110

is inserted may be smaller than a width of the collapsible form 110 in the deployed arrangement, such that when the collapsible form 110 (e.g., in the deployed arrangement) is disposed between the pair of adjacent form sections 120, the collapsible form 110 and the pair of adjacent form sections 120 bear or exert a force on one another, for example, such that relative positions of the collapsible form 110 and the form sections 120 are maintained.

[0054] Referring to FIG. 8, a form section 120 is illustrated in accordance with one example of the present disclosure. As shown in FIG. 8, the form section 120 may include a frame 810 and a form section panel 820. The form section panel 820 may be configured to define (e.g., at least a portion of) a boundary of a confined space in which a building material is poured and/or (e.g., at least partially) cured. According to some examples, the form section panel 820 may have a planar face configured to define a boundary of the confined space. According to some examples, the form section panel 820 may be composed of wood. For example, the form section panel 820 may be composed of plywood (e.g., one or more layers of wood glued to one another). According to other examples, the form section panel 820 may be composed of a metal or metal alloy. For example, the form section panel may be composed of aluminum, an aluminum alloy, steel, or another metal.

[0055] The frame 810 may be disposed behind the form section panel 820, for example, outside of the confined space. The frame 810 may be coupled to the form section panel 820. The frame 810 may be configured to support the form section panel 820. According to some examples, as shown in FIG. 8, the frame 810 may be disposed along a perimeter or periphery of the form section panel 820. The frame 810 may be comprised of a plurality of linear members or beams 811 coupled to one another. According to some examples, as shown in FIG. 8, a beam 811 of the frame 810 may extend along each side of the form section panel 820. According to some examples, the frame 810 may further include one or more (e.g., horizontal, vertical, diagonal) beams 811 extending between two of the beams 811 extending along respective sides of the form section panel 820. According to some examples, the frame 810 (e.g., the beams 811 included in the frame 810) may be composed of a metal or metal alloy. For example, the frame 810 (e.g., beam 811) may be composed of aluminum, an aluminum alloy, steel, or another metal. According to other examples, the frame 810 (e.g., beams 811) may be composed of wood.

[0056] The size and shape of the form section 120 (e.g., the frame 810 and form section panel 820) may vary. According to some examples, the form section 120 may have a rectangular (e.g., square) shape. According to other examples, the form section 120 may have another (e.g., polygonal) shape. A height of the form section 120 may vary. For example, the form section 120 may be 3 feet, 4 feet, 5 feet, 6 feet, 7 feet, 8 feet, 9 feet, or more than 9 feet tall.

[0057] In some examples, the collapsible form 110 may be (e.g., removably) coupled to one or more form sections 120. For example, each outer flange 220 may be (e.g., removably) coupled to a respective one of the pair of adjacent form sections 120.

[0058] An outer flange 220 of the collapsible form 110 and a form section 120 (e.g., directly) adjacent to the outer flange 220 may be coupled to one another. For example, as shown in FIGS. 2, 3, and 7, each outer flange 220 may include one or more coupling holes 223 extending therethrough. Addi-

tionally, as shown in FIG. 8, the form section 120, for example, the frame 810, may include one or more form section coupling holes 813 extending therethrough. As shown in FIG. 7, when the collapsible form 110 is disposed between a pair of adjacent form sections 120 and in the deployed arrangement, one or more form coupling holes 223 extending through an outer flange 220 may align with the one or more form section coupling holes 813 extending through the adjacent form section 120 and a fastener may be inserted through the aligned holes, coupling the collapsible form 110 and the form section 120.

[0059] For example, the fastener 900, illustrated in FIG. 9, may be inserted through the aligned holes, coupling the collapsible form 110 and the form section 120. As shown in FIG. 9, the fastener 900 may include a body 910 and a head 920 disposed at an end of the body 910. The fastener 900 may be inserted through an aligned form coupling hole 223 and form section coupling hole 813, for example, until the head 920 of the fastener 900 abuts one of the outer flange 220 of the collapsible form 110 or the frame 810 of the form section 120. According to other examples, another fastener, such as, a pin, bolt, rivet, or the like, may be inserted through the aligned holes, coupling the collapsible form 110 and the form section 120.

[0060] FIG. 10 illustrates a plurality of waler brackets 130 coupled to a collapsible form 110 in accordance with one example of the present disclosure. In some examples, the plurality of waler brackets 130 may be coupled to the collapsible form 110 in pairs. As shown in FIG. 1, each pair of waler brackets 130 may be configured to support or hold a rigid member 140 disposed between the pair of waler brackets 130 and the collapsible form 110. According to the present disclosure, one (e.g., a first one) of the pair of waler brackets 130 may be (e.g., directly) coupled to one (e.g., a first one) of the pair of outer flanges 220 and another (e.g., a second one) of the pair of waler brackets 130 may be (e.g., directly) coupled to another one (e.g., second one) of the pair of outer flanges 220. As described above, when the rigid member 140 is disposed between the collapsible form 110 and the pair of waler brackets 130, the rigid member 140 may prevent the collapsible form 110 from moving from the deployed arrangement.

[0061] Referring to FIG. 11, a waler bracket 130 is illustrated in accordance with one example of the present disclosure. As shown in FIG. 11, the waler bracket 130 may include a waler bracket body or a body 1110 and a waler bracket pin or a pin 1120. The body 1110 may be (e.g., directly, removably) coupled to the collapsible form, for example, collapsible form 110 illustrated in FIG. 2. The pin 1120 may be (e.g., directly, removably) coupled to the body 1110.

[0062] As shown in FIG. 11, the body 1110 may include a first or upper shelf 1111 and a second or lower shelf 1112. In accordance with some example, the upper shelf 1111 may define an upper surface of the waler bracket 130 configured to (e.g., directly) support a rigid member, for example, the rigid member 140 illustrated in FIG. 1. As shown in FIG. 11, the upper shelf 1111 may include a tab 1113. The tab 1113 may be inserted through a hole included in a collapsible form (e.g., collapsible form 110) and/or a form section (e.g., form section 120) removably coupling the waler bracket 130 to the collapsible form and/or the form section.

[0063] Returning to FIG. 2, the collapsible form 110 may include one or more waler bracket mounting holes 225. For

example, as shown in FIG. 2, one or more waler bracket mounting holes 225 may extend through each of the outer flanges 220. The one or more waler bracket mounting holes 225 may be configured to receive a tab of the waler bracket, for example, tab 1113 illustrated in FIG. 11, such that the waler bracket may be removably coupled to the collapsible form 110.

[0064] In accordance with some examples, the waler bracket mounting holes 225 and the form coupling holes 223 may have the same size and/or shape, such that, the waler bracket mounting holes 225 may be configured to receive a fastener, for example, fastener 900 illustrated in FIG. 9, for coupling the collapsible form 110 to an adjacent form section and/or the form coupling holes 223 may be configured to receive a tab for coupling a waler bracket to the collapsible form 110. In accordance, with other examples, the waler bracket mounting holes 225 and the form coupling holes 223 may have different size and/or shapes, such that the waler bracket mounting holes 225 are not capable of receiving a fastener, for example, fastener 900 illustrated in FIG. 9, for coupling the collapsible form to an adjacent form section and the form coupling holes 223 are not capable of receiving a tab for coupling a waler bracket to the collapsible form 110.

[0065] Returning to FIG. 11, the tab 1113 may be configured to be removably inserted into a waler bracket mounting hole directly coupling the body 1110 to the collapsible form. In accordance with some examples, the waler bracket 130 may be (e.g., removably, directly) coupled to both a collapsible form and a form section (e.g., adjacent to the collapsible form). For example, a tab 1113 of the waler bracket 130 may extend through both a waler bracket mounting hole (e.g., 225 illustrated in FIG. 2) of the collapsible form and a form section coupling hole (e.g., 813 illustrated in FIG. 8, which are aligned).

[0066] The lower shelf 1112 may include a wall 1114 extending (e.g., downward) from the lower shelf 1112. The wall 1114 may be configured to abut an outer flange of the collapsible form when the waler bracket 130 is coupled to the collapsible form. For example, as shown in FIG. 10, when the tab 1113 of the waler bracket 130 extends through a waler bracket mounting hole 225, the wall 1114 of the waler bracket 130 may contact an outer flange 220 of the collapsible form 110. In some examples, the wall 1114 may also contact the frame of a form section (e.g., directly) adjacent to the collapsible form 110.

[0067] Returning to FIG. 11, the pin 1120 may be removably (e.g., directly) coupled to the body 1110 of the waler bracket 130. As shown in FIG. 11, in some examples, the pin 1120 may have a substantially “U” shape including a web 1121 and a pair of legs 1122 extending (e.g., in the same direction) from opposite ends of the web 1121. In accordance with some examples of the present disclosure, when the pin 1120 is coupled to the body 1110, the web 1121 of the pin 1120 may be configured to abut a rigid member, for example, rigid member 140 illustrated in FIG. 1, disposed between the waler bracket 130 (e.g., the pin 1120) and the collapsible form (e.g., a locking brace and/or outer flanges).

[0068] In accordance with some examples, the pin 1120 may be inserted through a slot or an upper slot 1115 extending through the upper shelf 1111 and/or a slot or lower slot 1116 extending through the lower shelf 1112. In accordance with some examples, the upper slot 1115 and/or lower slot 1116 may have a shape corresponding to the pin 1120.

For example, as shown in FIG. 11, the upper slot 1115 may include a web portion 1118 configured to receive the web 1121 of the pin 1120 and two leg portions 1119 configured to receive respective legs 1122 of the pin 1120.

[0069] Returning to FIG. 1, the forming system 100 further includes a rigid member 140. The rigid member 140 may be configured to be provided (e.g., placed, positioned) between a pair of waler brackets 130 and the collapsible form 110. The rigid member 140 may be configured to abut the waler bracket pins, for example, waler bracket pins 1120 illustrated in FIG. 11, and the one or more locking braces, for example, locking braces 260 illustrated in FIG. 2, when the rigid member 140 is disposed between the pair of waler brackets 130 and the collapsible form 110. According to some examples, as shown in FIG. 1, the rigid member 140 may contact an end of each of the pair of outer flanges of the collapsible form 110 when the rigid member is disposed between the pair of waler brackets 130 and the collapsible form 110. The rigid member 140 (e.g., when disposed between the pair of waler brackets 130 and the collapsible form) may be configured to prevent the collapsible form 110 from moving from the deployed arrangement to the collapsed arrangement. The rigid member 140 may be rigid, such that, the rigid member 140 does not deform in response to a force applied to the rigid member 140 by the waler brackets 130 and/or collapsible form 110.

[0070] The size and/or shape of the rigid member 140 may vary. In some examples, the size and/or shape of the rigid member 140 may vary in accordance with a size and/or shape of the pair of waler brackets 130 coupled to the collapsible form 110. For example, the rigid member 140 may have a size and/or shape (e.g., about at least one axis) corresponding to a distance between the pair of waler bracket pins and the collapsible form 110 (e.g., the one or more locking braces and/or a back end of the outer flanges). According to some examples, the rigid member 140 may be composed of wood. For example, the rigid member 140 may be a piece of lumber (e.g., a 2"×4" piece of lumber, 1.5"×3.5" piece of lumber, any other size piece of lumber). According to other examples, the rigid member 140 may be composed of another material, for example, a metal or metal alloy, a plastic or plastics, or the like.

[0071] Referring to FIG. 12, a flowchart 1200 for using a collapsible form, for example, collapsible form 110 illustrated in FIG. 2, and/or a forming system, for example, forming system 100 illustrated in FIG. 1, is illustrated in accordance with one example of the present disclosure. Additional, different, or fewer acts may be provided.

[0072] At 1201, a collapsible form may be positioned or placed between a pair of adjacent form sections. For example, a user (e.g., person) may place the collapsible form between a pair of adjacent form sections. In accordance with some examples, the collapsible form may be in the collapsed arrangement when the collapsible form is placed between the pair of adjacent form sections. Accordingly, a space (e.g., opening, gap) may be disposed between the collapsible form and one or both of the form sections when the collapsible form is in the collapsed arrangement.

[0073] At 1203, the collapsible form may transition or move from the collapsed arrangement to the deployed arrangement. For example, a user may move the collapsible form from the collapsed arrangement to the deployed arrangement using a handle of the collapsible form. In accordance with some examples, as the collapsible form is

moved to the deployed arrangement, the collapsible form may engage (e.g., abut, contact) the pair of adjacent form sections. For example, the outer flanges of the collapsible form may be brought into contact with a respective one of the pair of adjacent form sections.

[0074] At **1205**, one or more fasteners, for example, fastener **900** illustrated in FIG. 9, may be inserted through aligned holes of the collapsible form and a respective one of the pair of adjacent form sections. For example, the fastener may be inserted through a hole, for example, a form coupling hole (e.g., **223** as illustrated in FIG. 2) extending through an outer flange of the collapsible form and a form section coupling hole (e.g., **813** as illustrated in FIG. 8), coupling the collapsible form to a respective one of the pair of adjacent form sections.

[0075] At **1207**, two or more waler brackets, for example, waler brackets **130** illustrated in FIG. 11, may be coupled to the collapsible form and/or the pair of adjacent form sections. For example, a tab of the waler bracket may be inserted through a waler bracket mounting hole extending through the collapsible form. In some examples, the tab may be inserted through a waler bracket mounting hole of the collapsible form and a form section coupling hole extending through the frame of a form section (e.g., directly) adjacent to the collapsible form, such that the waler bracket is (e.g., directly, removably) coupled to both the collapsible form and form section.

[0076] Additionally, the pins, for example, pins **1120** illustrated in FIG. 11, may be inserted into the body, for example, body **1110** illustrated in FIG. 11, of the waler bracket at **1207**.

[0077] At **1209**, a rigid member, for example, rigid member **140** illustrated in FIG. 1, may be provided between a pair of waler brackets and the collapsible form. For example, the rigid member may be provided (e.g., placed, positioned) between the respective pins of a pair of waler brackets and the collapsible form (e.g., locking braces and/or outer flanges of the collapsible form).

[0078] At **1211**, the building material may be poured into and (e.g., at least partially) cured within the confined space defined by the collapsible form and the plurality of form sections. As described above, according to some examples, concrete may be poured and at least partially cured within a confined space defined by the collapsible form (disposed in the deployed arrangement) and one or more form sections. In accordance with other examples, plaster, asphalt, or another building material may be poured and at least partially cured within the confined space.

[0079] At **1213**, the pins of the waler bracket may be removed from the body of the waler bracket. According to the present disclosure, when the building material (e.g., concrete) is poured and (e.g., at least partially) cured within the confined space, a weight or pressure of the building material may exert an outward force on the collapsible form, for example, the inner panels of the collapsible form. In some examples, the weight or pressure of the building material may cause the collapsible form, rigid member, and waler bracket to be wedged tightly together. In accordance with some examples, the pins, for example, pins **1120** illustrated in FIG. 11, may first be separated or decoupled from the pair of waler brackets, such that a rigid member is free to move away from the collapsible form. In some examples, a force may be applied to (e.g., a bottom of) the pins, such that the pins may be pushed upward and out of

slots (e.g., upper and/or lower slots) included in the body of the waler bracket. For example, a hammer or another tool may be used to strike a bottom of the pins, pushing the pins upward and out of their respective slots in the body of the waler bracket.

[0080] At **1215**, the rigid member may be removed or separated from the waler brackets and/or the collapsible form. According to some examples, as described above, the pins of the waler bracket may be removed before the rigid member is separated from the waler brackets and/or the collapsible form, in these examples, the rigid member may no longer be disposed between the collapsible form and the pins of the waler bracket such that the rigid member is free to simply be moved away from the collapsible form and the body of the waler brackets. In accordance with other examples, the waler bracket pins may still be coupled to the waler bracket body when the rigid member is separated from the collapsible form and the waler brackets. In these examples, the rigid member may be moved upward so as to be separated from the waler brackets and/or collapsible form.

[0081] At **1217**, the fasteners may be removed from aligned holes of the collapsible form and the pair of adjacent form sections. After removal of the fasteners the collapsible form and adjacent form sections may be free move relative to (e.g., be separated from) one another.

[0082] At **1219**, the collapsible form may move or transition from the deployed arrangement to the collapsed arrangement. For example, a user may move the collapsible form from the deployed arrangement to the collapsed arrangement. According to the present disclosure, movement of the collapsible form from the deployed arrangement to the collapsed arrangement may facilitate removal or separation of the collapsible form from the pair of adjacent form sections and/or a structure formed by the building material. In accordance with some examples, the collapsible form may be moved to the collapsed arrangement before being moved from between the pair of adjacent form sections. In accordance with other examples, the collapsible form may be moved to the collapsed arrangement as it is being removed from between the pair of adjacent form sections. In accordance with one example of the present disclosure, the collapsible form may be provided as a 12" wide "U" shaped panel. In other examples, the collapsible form may be provided as a 4" wide "U" shaped panel. In accordance with some examples of the present disclosure, the collapsible form may be 3', 4', 5', 6', 8', or 9' tall.

[0083] According to some examples of the present disclosure, the collapsible form may be comprised of a metal or metal alloy. In some examples, the collapsible form may be comprised of steel. In other examples, the collapsible form may be comprised of aluminum or an aluminum alloy. In some examples, the collapsible form and the form sections (e.g., form system) the collapsible form is intended to be used with may be comprised of the same material. For examples, a collapsible form configured to be used with steel plywood form sections may be comprised of steel. Similarly, a collapsible form configured to be used with aluminum form sections may be comprised of aluminum or aluminum alloy.

[0084] According to the present disclosure, the collapsible form may be compatible with (e.g., configured to be coupled to) any currently existing or later developed steel plywood forms or forming systems. For example, the collapsible form

may be compatible with EMI steel plywood forms, Symons steel plywood forms, SureBuilt Handset steel plywood forms, and the like.

[0085] According to the present disclosure, the collapsible form may be compatible with (e.g., configured to be coupled to) any currently existing or later developed aluminum forming systems. For example, the collapsible form may be compatible with EMI aluminum forms, Western Forms aluminum forms, Precise Forms aluminum forms, and the like.

[0086] The number of waler brackets coupled to the collapsible form (and thus a number of waler bracket mounting holes) may vary. An even number of waler brackets may be coupled to the collapsible form and an equal number of waler brackets may be coupled to each of the pair of outer flanges. The number of rigid members coupled to the collapsible form may correspond to the number of waler brackets coupled to each of the outer flanges. The number of waler brackets coupled to each of the pair outer flanges may vary in accordance with a height of the collapsible form; taller collapsible forms may include more waler brackets.

[0087] Two or more collapsible forms may be used for a single concrete pour or casting. Specifically, two or more collapsible forms may define a boundary of the same confined space in which concrete is poured. In some examples, the two or more collapsible forms may define opposite boundaries of a confined space in which concrete is poured. In some examples, the two or more collapsible forms may define a boundary of different walls included in a confined space in which concrete is poured.

[0088] In accordance with some examples of the present disclosure, each of the pair of outer flanges may include one or more waler bracket mounting holes configured to receive a waler bracket and one or more form coupling holes configured to receive a fastener for coupling the collapsible form to an adjacent form section. In some examples, the waler bracket mounting holes and form coupling holes may have the same shape and/or size such that the waler bracket mounting holes and form coupling holes both be configured to receive a waler bracket and a fastener.

[0089] While the present invention has been described with reference to specific examples, which are intended to be illustrative only and not to be limiting of the invention, it will be apparent to those of ordinary skill in the art that changes, additions and/or deletions may be made to the disclosed embodiments without departing from the spirit and scope of the invention.

[0090] The foregoing description is given for clarity of understanding only, and no unnecessary limitations should be understood therefrom, as modifications within the scope of the invention may be apparent to those having ordinary skill in the art.

What is claimed is:

1. A collapsible concrete form comprising:

a pair of inner panels hingedly coupled to one another;
a pair of outer flanges configured to abut a pair of adjacent form sections, such that the pair of outer flanges and the pair of inner panels are disposed between the pair of adjacent form sections, a first one of the pair of outer flanges hingedly coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges hingedly coupled to a second one of the pair inner panels; and

a handle protruding from one of the pair of inner panels or a hinge coupling the pair of inner panels, the handle operable to move the pair of inner panels and the pair of outer flanges between a deployed arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner panels are askew.

2. The collapsible concrete form of claim 1, further comprising:

a locking brace protruding from one of the pair of inner panels.

3. The collapsible concrete form of claim 2, wherein the locking brace includes:

a pair of projections extending from one of the pair of inner panels; and

a bearing panel extending between the pair of projections and disposed parallel to one or the pair of inner panels.

4. The collapsible concrete form of claim 2, further comprising:

a pair of waler bracket mounting holes configured to receive a waler bracket, a first one of the pair of waler bracket mounting holes extending through the first one of the pair of outer flanges and a second one of the pair of waler bracket mounting holes extending through the second one of the pair of outer flanges,

wherein the pair of waler bracket mounting holes are configured to receive a pair of waler brackets and in the deployed arrangement, the pair of waler brackets are configured to support a rigid member disposed between the pair of waler brackets and the locking brace and the pair of waler brackets and the locking brace are configured to abut the rigid member, preventing the pair of inner panels and the pair of outer flanges from moving from the deployed arrangement.

5. The collapsible concrete form of claim 1, further comprising:

a pair of side hinges; and

wherein a first one of the pair of outer flanges and a first one of the pair of inner panels collectively form a first one of the pair of side hinges,

wherein a second one of the pair of outer flanges and a second one of the pair of inner panels collectively form a second one of the pair of side hinges, and

wherein the pair of inner panels collectively form the hinge coupling the pair of inner panels.

6. The collapsible concrete form of claim 1, wherein in the deployed arrangement, the pair of inner panels are configured to define a planar boundary of confined space in which concrete is poured.

7. The collapsible concrete form of claim 1, further comprising a form coupling hole extending through one of the pair of outer flanges, the form coupling hole configured to receive a fastener for coupling a respective outer flange of the pair of outer flanges to an adjacent form section of the pair of adjacent form sections.

8. A concrete forming system comprising:

a collapsible form including:

a pair of inner panels hingedly coupled to one another;

a pair of outer flanges configured to abut an adjacent form section, a first one of the pair of outer flanges hingedly coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges hingedly coupled to a second one of the pair inner panels; and

a pair of waler bracket mounting holes configured to receive a waler bracket, a first one of the pair of waler bracket mounting holes extending through the first one of the pair of outer flanges and a second one of the pair of waler bracket mounting holes extending through the second one of the pair of outer flanges; and

a pair of waler brackets configured to be inserted into the pair of waler bracket mounting holes,

wherein the pair of inner panels and the pair of outer flanges are configured to move between a deployed arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner panels are askew.

9. The concrete forming system of claim 8, further comprising:

- a locking brace protruding from one of the pair of inner panels; and
- a rigid member, wherein in the deployed arrangement the pair of waler brackets are configured to support the rigid member disposed between the pair of waler brackets and the locking brace,

wherein when the rigid member is disposed between the pair of waler brackets and the locking brace, the pair of waler brackets and the locking brace abut the rigid member, preventing the pair of inner walls and the pair of outer flanges from moving from the deployed arrangement.

10. The concrete forming system of claim 9, wherein the locking brace includes:

- a pair of projections extending from one of the pair of inner panels; and
- a bearing panel extending between the pair of projections, the bearing panel disposed parallel to one of the pair of inner panels and configured to abut a rigid member disposed between the pair of waler brackets and the locking brace.

11. The concrete forming system of claim 9, wherein when the rigid member is disposed between the pair of waler brackets and the locking brace, the pair of outer flanges are configured to abut the rigid member.

12. The concrete forming system of claim 8, further comprising:

- a handle protruding from one of the pair of inner panels or a hinge coupling the pair of inner panels, the handle operable to move the pair of inner panels and the pair of outer flanges between the deployed arrangement and the collapsed arrangement.

13. The concrete forming system of claim 8, further comprising:

- a form coupling hole extending through one of the pair of outer flanges;
- a form section including:
 - a form section panel;
 - a frame coupled to the form section panel and disposed along a perimeter of the form section panel; and
 - a form section coupling hole extending through the frame; and
- a fastener configured to be inserted through the form coupling hole and the form section coupling hole, coupling the collapsible form and the form section.

14. The concrete forming system of claim 13, wherein when the collapsible form and the form section are coupled, the pair of inner walls and the form section panel are

collinear and collectively define planar boundary of a confined space in which concrete is poured.

15. The concrete forming system of claim 13, wherein the form section coupling hole comprises a pair of form section coupling holes,

- wherein the fastener is configured to be inserted through the form coupling hole and one of the pair of form section coupling holes, coupling the collapsible form and the form section, and

- wherein one of the pair of waler brackets is configured to be inserted through one of the pair of waler bracket mounting holes and an other of the pair of form section coupling holes.

16. A collapsible concrete form comprising:

- an inner portion including a pair of inner panels pivotably coupled to one another, the inner portion configured to move between a linear arrangement in which the pair of inner panels are collinear and a collapsed arrangement in which the pair of inner walls are askew; and

- a pair of outer flanges configured to abut a pair of adjacent form sections, such that the pair of outer flanges and the inner portion are disposed between the pair of adjacent form sections, a first one of the pair of outer flanges pivotably coupled to a first one of the pair of inner panels and a second one of the pair of outer flanges pivotably coupled to a second one of the pair inner panels.

17. The collapsible concrete form of claim 16, further comprising:

- a locking brace protruding from one of the pair of inner panels; and

- a pair of waler bracket mounting holes configured to receive a pair of waler brackets, a first one of the pair of waler bracket mounting holes extending through the first one of the pair of outer flanges and a second one of the pair of waler bracket mounting holes extending through the second one of the pair of outer flanges,

- wherein in the linear arrangement, the pair of waler brackets are configured to support a rigid member disposed between the pair of waler brackets and the locking brace and the pair of waler brackets and the locking brace are configured to abut the rigid member, preventing the pair of inner panels from moving from the linear arrangement.

18. The collapsible concrete form of claim 17, wherein the locking brace includes:

- a pair of projections extending from one of the pair of inner panels; and

- a bearing panel extending between the pair of projections and disposed parallel to one or the pair of inner panels, the bearing panel configured to abut a rigid member disposed between the pair of waler brackets and the locking brace.

19. The collapsible concrete form of claim 17, further comprising:

- a handle protruding from one of the pair of inner panels or a hinge coupling the pair of inner panels, the handle operable to move the pair of inner panels between the linear arrangement and the collapsed arrangement when the rigid member is not disposed between the pair of waler brackets and the locking brace.

20. The collapsible concrete form of claim 17, wherein when the inner portion is in the linear arrangement, the pair of inner walls define a planar boundary of confined space in which concrete is poured.

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